GERGELY FLAMICH

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PERSONAL STATEMENT

Since August 2025, I have been an Imperial College Research Fellow at Imperial College London. I have deep expertise in machine learning with strong practical and theoretical research and coding skills. My current work focuses on **neural data compression** using generative models such as diffusion models and variational autoencoders, and related **information theory**, mainly **relative entropy coding**.

PROFESSIONAL EXPERIENCE

Imperial College Research Fellow (Research Associate until July 2025) — March 2025 - Now
Host: Dr Deniz Gündüz — *Imperial College London*

- Characterised the absolute best theoretically achievable data compression using a new fundamental measure of information called the *functional information* (a generalisation of entropy). Paper in preparation for IEEE Transactions on Information Theory.
- Developed the first data compression algorithm that has **provably optimal compression performance**. Implementation and analysis mainly in `Jax`. Paper to appear at IEEE ITW 2026.
- Implemented scalable, privacy-preserving image compression algorithm using diffusion models that compresses data $\times$**20-40 more** than competing approaches. Implementation mostly in `PyTorch`. Workshop paper to appear at ICML 2026, full paper in preparation for ICLR 2027.
- Co-organised the Learn to Compress & Compress to Learn Workshop at the International Symposium on Information Theory (ISIT) 2025; currently assisting in organisation of the 2026 rendition.
- Was elected as one of 16 new research fellows at Imperial College in August 2025, based on an independently written research proposal. As the University's website states, the fellowship 'is designed for exceptional early career researchers who are prepared to step into independence.' The fellowship covers salary as well as £30,000 for equipment and travel expenses.
- Since start of postdoc, 3 papers to appear at information theory and ML venues, of which one received spotlight award; 3 submitted for peer review. Assistant supervisor to first-year PhD student (Ö. Güner) and supervising an MPhil research project at Cambridge on video compression with diffusion models.

Student Researcher: Neural Machine Translation — Dec 2024 - Feb 2025
Hosts: Dr David Vilar, Dr Jan-Thorsten Peter and Dr Markus Freitag — *Google Berlin*

- Implemented discrete diffusion models in `Jax` and `T5X` to speed up neural machine translation.
- Developed theory of the accuracy-naturalness tradeoff in translation and demonstrated that current methodology for automatic evaluation of translation systems is inadequate and needs to be updated.
- Work resulted in a paper published and presented at the Conference on Language Modeling 2025.

Student Researcher: Relative Entropy Coding for Practical Data Compression — July 2022 - Dec 2022
Host: Dr Lucas Theis — *Google Brain, London*

- Developed data compression algorithm that is significantly faster than previous comparable methods. Implementation in `TensorFlow`.
- Published and presented work at IEEE International Symposium on Information Theory 2023.
- Contributed the noncentral χ^2 distribution to the open-source TensorFlow Probability package.

PhD in Advanced Machine Learning — Oct 2020 - July 2025
Supervisor: Dr José Miguel Hernández-Lobato — *University of Cambridge*

- Developed new theoretical foundations for lossy data compression using randomised codes.
- Applied the theoretical findings to develop lossy data compression algorithms using generative models, such as variational autoencoders and Bayesian implicit neural representations.

- Published 14 papers (first author on 8) with 11 coauthors at top AI/ML and information theory conferences, such as NeurIPS, ICLR, ICML, COLM and ISIT. Received **spotlight award (top 10% accepted papers)** at NeurIPS for work on efficient data compression with Bayesian INRs. Proposed and supervised three MPhil research projects and two undergraduate research projects.

Research Assistant — Oct 2019 - July 2020
Supervisor: Dr José Miguel Hernández-Lobato — *University of Cambridge*

- Implemented and optimized black-box optimization pipeline to find optimal hardware designs in collaboration with researchers at Harvard and Intel. Implementation in `TensorFlow` and `GPFlow`.
- Our method consistently found better configurations faster than standard methods used in industry.

EDUCATION

PhD in Advanced Machine Learning — Oct 2020 - July 2025
Supervisor: *Dr José Miguel Hernández-Lobato* — *University of Cambridge*

MPhil in Machine Learning and Machine Intelligence — Oct 2018 - Oct 2019
*Graduated with **Commendation** (roughly top 20% of class)* — *University of Cambridge*

BSc Joint Honours in Mathematics and Computer Science — Oct 2014 - Jun 2018
*Graduated as **Valedictorian** in Computer Science, with First Class Honours* — *University of St Andrews*

ACADEMIC ACHIEVEMENTS

2026	Spotlight Paper Award	Learn to Compress Workshop @ ISIT 2026
2025	Imperial College Research Fellowship	Equivalent to Leverhulme Trust & Wellcome Trust Early Career Fellowships. Success rate: 17.98% of applicants (2025).
2025	Runner-up for the Jack Keil Wolf student paper award	ISIT 2025
2024	Spotlight Paper Award	Learn to Compress Workshop @ ISIT 2024
2023	Spotlight Paper Award	NeurIPS 2023, a top venue for machine learning research.
2021	Finalist	Qualcomm Innovation Fellowship

TECHNICAL SKILLS

Programming Languages	Python, Javascript, Java, Haskell, Matlab, C, C++, LaTeX
ML Frameworks & APIs	Jax AI Stack, TensorFlow, TensorFlow Probability, SciPy

SELECTED PUBLICATIONS

• **G. Flamich** and S. Hill (2026). Singular relative entropy coding with bits-back rejection sampling. To appear in *Learn to Compress & Compress to Learn Workshop @ ISIT 2026*. Received **Spotlight award**.
• **G. Flamich**, S. M. Sriramu and A. B. Wagner (2025). The Redundancy of Non-Singular Channel Simulation. In *ISIT 2025*. Runner-up to the Jack Keil Wolf student paper award (**top 6 student paper**)
• J. He†, **G. Flamich**†, Z. Guo, J. M. Hernández-Lobato. RECOMBINER: Robust and Enhanced Compression with Bayesian Implicit Neural Representations. In *ICLR 2024*.
• **G. Flamich**. Greedy Poisson Rejection Sampling. In *NeurIPS 2023*.
• **G. Flamich**†, Z. Guo†, J. He, Z. Chen, J. M. Hernández-Lobato. Compression with Bayesian Implicit Neural Representations. In *NeurIPS 2023*. Received **Spotlight award** (top 10% of accepted papers).
• **G. Flamich**†, S. Markou†, J. M. Hernández-Lobato. Fast Relative Entropy Coding with A* coding. In *ICML 2022*.
• **G. Flamich**†, M. Havasi†, J. M. Hernández-Lobato. Compressing Images by Encoding Their Latent Representations with Relative Entropy Coding. In *NeurIPS 2020*.
† equal contribution.